

Tyre Degradation & Pit Window Optimization Model

Simplified race strategy model in MATLAB

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Introduction

Tyre degradation is one of the most influential factors in motorsport strategy. As a tyre completes more laps, its grip level, thermal behaviour and consistency progressively change, affecting the driver's ability to maintain competitive lap times. For this reason, tyre performance is not only a mechanical or vehicle dynamics issue, but also a strategic variable that directly influences stint length, pit stop timing and overall race execution.

In categories such as Formula 1, the decision of when to stop is rarely based on a single factor. Teams must consider the balance between tyre ageing, fuel load reduction, compound performance, pit stop time loss and the expected pace over the remaining laps. A softer compound may provide better initial performance, but it can also suffer from higher degradation. A harder compound may be slower at the beginning of the stint, but it can offer greater stability over a longer distance. Therefore, the fastest compound in isolation is not always the best strategic option over a complete race distance.

This project develops a simplified MATLAB-based model to analyse tyre degradation and pit stop strategy. The model simulates the behaviour of three tyre compounds — Soft, Medium and Hard — using assumed parameters for base pace, non-linear degradation, fuel burn-off and tyre operating limits. It then compares different one-stop race strategies and identifies the pit stop lap that minimizes total race time.

The purpose of this project is not to replicate the exact behaviour of a real Formula 1 or MotoGP race, since real tyre performance depends on confidential data, track conditions, temperature, driver inputs, traffic, setup and many other variables. Instead, the goal is to build a clear and structured engineering model that captures the strategic logic behind tyre degradation and pit stop optimization. By doing so, the project demonstrates how simplified assumptions can be transformed into a quantitative decision-making tool for motorsport performance analysis.

1. Project Objective

The main objective of this project is to develop a simplified race strategy model capable of evaluating how tyre degradation, fuel burn-off and compound selection affect the optimal pit stop window in a one-stop race scenario.

More specifically, the project aims to:

1. **Model tyre degradation behaviour** for three different compounds: Soft, Medium and Hard. Each compound is defined by an initial pace, a linear degradation coefficient and a non-linear degradation coefficient.
2. **Include the effect of fuel burn-off**, assuming that lap time improves progressively as the car becomes lighter during the race.
3. **Introduce fuel-load sensitivity**, meaning that tyre degradation is affected by the amount of fuel carried by the car. This prevents unrealistic symmetry between strategies such as Soft-Medium and Medium-Soft, since a tyre used at the start of the race is exposed to higher load and stress than the same tyre used later.
4. **Apply a tyre cliff penalty** as a modelling assumption to discourage excessively long stints beyond the estimated operating window of each compound. This penalty is not intended to reproduce a specific regulation, but to represent the possible sharp drop in performance when a tyre is used beyond its effective range.
5. **Compare different one-stop strategies** using valid compound combinations, where the second stint must use a different compound from the starting tyre.
6. **Identify the optimal pit stop lap** for each strategy by testing multiple possible pit stop laps and selecting the one that produces the lowest total race time.
7. **Determine the fastest technically plausible strategy** under the assumptions of the model, while also discussing the limitations of the simulation.

The final goal is to create a compact but meaningful motorsport engineering project that combines modelling, data analysis, MATLAB simulation and strategic interpretation. This project aims to show the ability to structure a performance problem, define assumptions, build a simulation and extract actionable conclusions from the results.

2. Model Assumptions

This project is based on a simplified race strategy model. The objective is not to reproduce the exact behaviour of a real Formula 1 or MotoGP race, but to create a structured simulation framework that captures the main strategic relationships between tyre degradation, fuel burn-off, compound selection and pit stop timing.

Several assumptions were defined in order to keep the model understandable. These assumptions allow the simulation to compare different one-stop strategies under the same conditions.

VARIABLE	ASSUMPTION
Race Length	50 laps
Strategy type	One-stop race strategy

Pit stop loss	22 seconds
Tyre compounds	Soft, Medium, Hard
Compound change	The second stint must use a different compound from the starting tyre
Tyre degradation	Non-linear degradation based on tyre age
Fuel burn-off	Lap time improves progressively as fuel mass decreases
Fuel-load sensitivity	Tyre degradation is higher when the car is heavier at the beginning of the race
Tyre cliff	Additional penalty is applied if a tyre exceeds its estimated operating window

The model uses assumed numerical parameters for each tyre compound. These values are not intended to represent official tyre supplier data or confidential team information. Instead, they are used to create a plausible comparison between compounds with different performance profiles.

3. Tyre Degradation Model

The tyre degradation model estimates lap time evolution as a function of tyre age, compound characteristics and race lap. The aim is to represent, in a simplified way, how tyre performance changes during a stint and how this affects the overall race strategy.

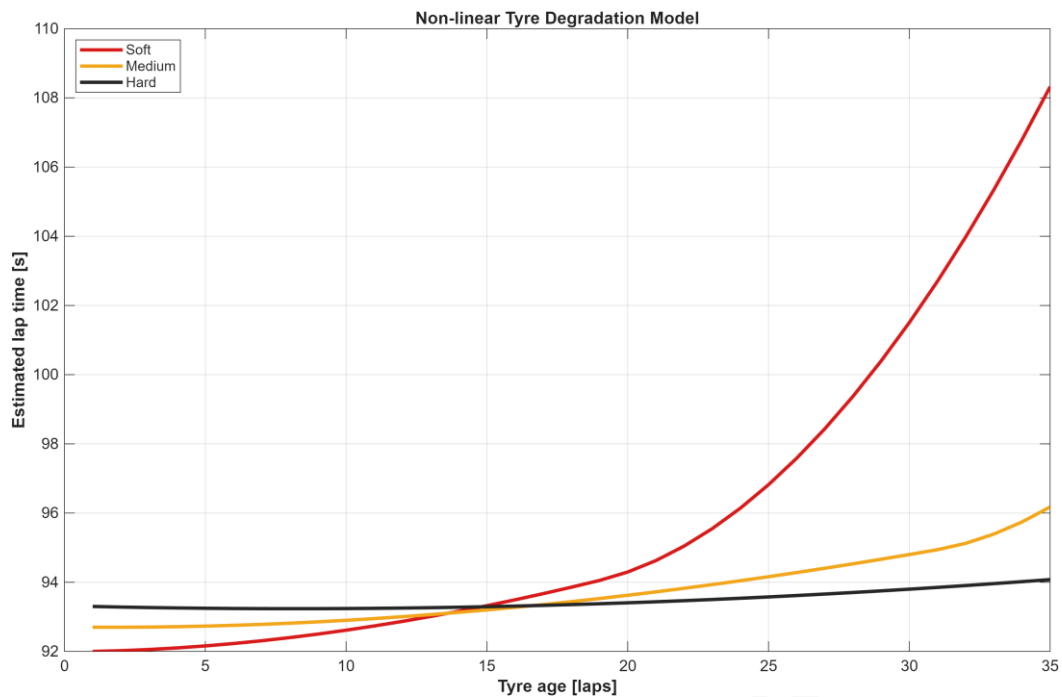
The model assumes that lap time is influenced by four main components:

- Lap Time = Base Pace + Tyre Degradation + Tyre Cliff Penalty – Fuel Burn-off Gain

The assumed compound parameters are shown below:

COMPOUND	BASE PACE	LINEAR DEGRADATION	NON-LINEAR DEGRADATION	STIMATED OPERATING WINDOW
Soft	92.00	0.035	0.0065	18 laps
Medium	92.70	0.025	0.0025	30 laps
Hard	93.30	0.015	0.0012	42 laps

These values are assumed parameters, not official tyre data. They are used to create a realistic performance.



4. Strategy Simulation Methodolgy

The strategy simulation is based on a simplified one-stop race scenario. Each strategy is composed of two tyre stints and one pit stop. To keep the model aligned with a realistic dry-race strategy framework, the second compound must be different from the starting compound.

For each valid strategy, the model tests different possible pit stop laps. In every case, it simulates the lap time evolution of the first stint, then resets tyre age after the pit stop and simulates the second stint with the new compound. The pit stop time loss is added to the selected pit lap, and the total race time is calculated.

The optimal pit stop lap is therefore defined as the lap that minimizes the total simulated race time for a given tyre strategy.

A simplified excerpt of the MATLAB logic used to calculate lap time is shown below:

```
tyreDegradation = fuelStressMultiplier * ...
    (compound.degLinear * tyreAge + compound.degQuad * tyreAge^2);

cliffPenalty = 0;

if tyreAge > compound.maxStint
    lapsOverLimit = tyreAge - compound.maxStint;
    cliffPenalty = 0.04 * lapsOverLimit^2;
end

lapTime = compound.basePace + tyreDegradation + cliffPenalty - fuelBurnOffGain;
```

This code shows how the model combines non-linear tyre degradation, fuel-load sensitivity, tyre cliff penalty and fuel burn-off to estimate lap time during each stint.

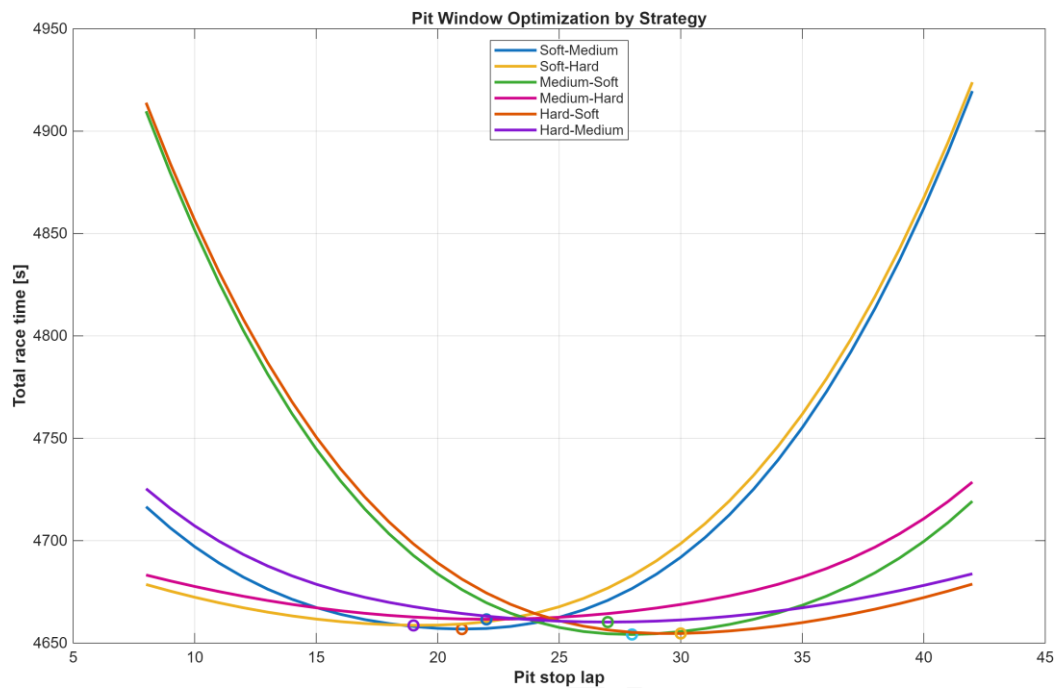


Figure 2. Pit window optimization across valid one-stop strategies.

The pit window optimization graph shows how total race time changes depending on the selected pit stop lap. Each curve represents a different tyre strategy. The lowest point of each curve indicates the optimal pit stop lap for that specific strategy.

5. Results

The simulation produced three main outputs: the pit window optimization for each valid strategy, the total race time comparison between strategies, and the lap time profile of the best-performing strategy. These results allow the model to move from tyre behaviour analysis to strategic decision-making.

5.1. Pit Window Optimization

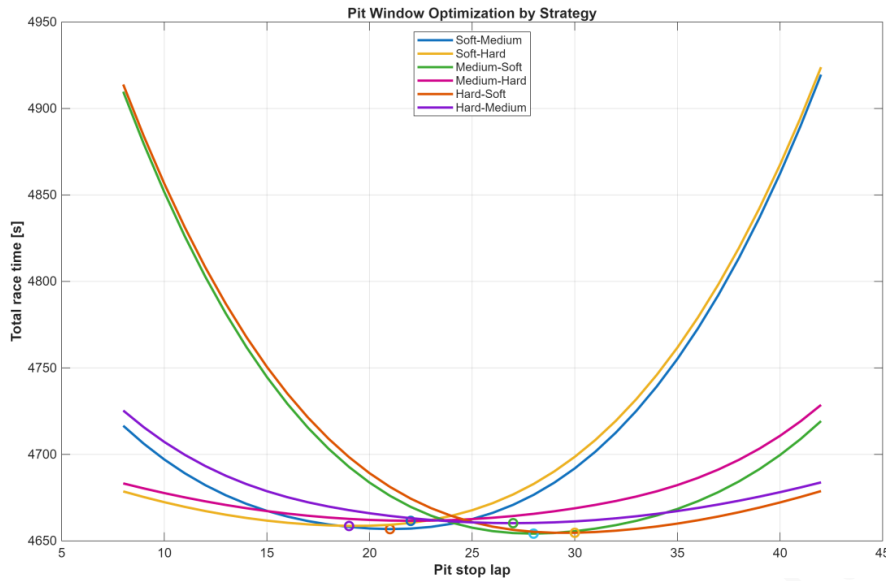


Figure 2. Pit window optimization across valid one-stop strategies.

The pit window optimization graph shows how total race time changes depending on the selected pit stop lap. Each curve represents a different one-stop strategy. The lowest point of each curve indicates the optimal pit stop lap for that specific compound combination.

The results show that the optimal pit stop lap is not fixed, but depends on the balance between the starting compound, the second compound, tyre degradation, fuel-load sensitivity and pit stop time loss. Strategies using softer compounds tend to require earlier pit stops due to their higher degradation, while harder compounds allow longer first stints but may sacrifice initial pace.

This graph is useful because it shows that pit stop timing is not simply a question of stopping as early or as late as possible. Instead, the optimal pit window appears when the time lost through tyre degradation becomes greater than the advantage of extending the current stint.

5.2. Total Race Time Comparison

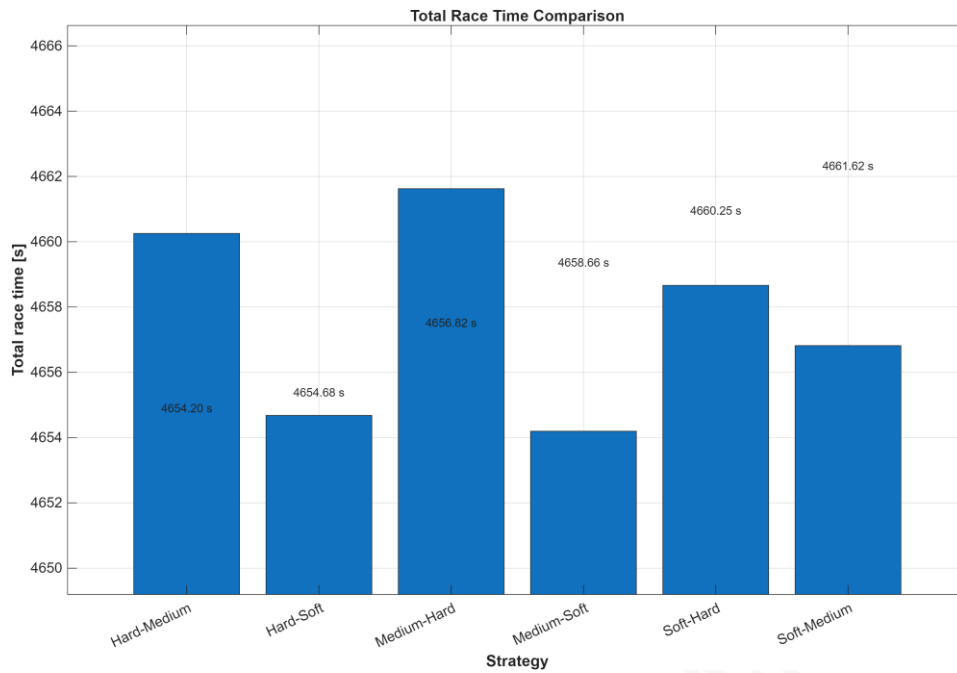


Figure 3. Strategy time loss relative to the best option.

Since the total race times between strategies are relatively close, the comparison is presented as a time loss relative to the fastest strategy. This makes it easier to visualize small but strategically relevant differences.

The best strategy is used as the reference point, while the remaining strategies are shown according to how much time they lose compared with the optimal option. Even if the absolute time differences are small, these gaps can be important in motorsport, where a few tenths per lap or a few seconds across a race distance can determine track position.

This comparison helps identify not only the fastest strategy, but also which alternatives remain competitive under the assumptions of the model.

5.3 Best Strategy Lap Time

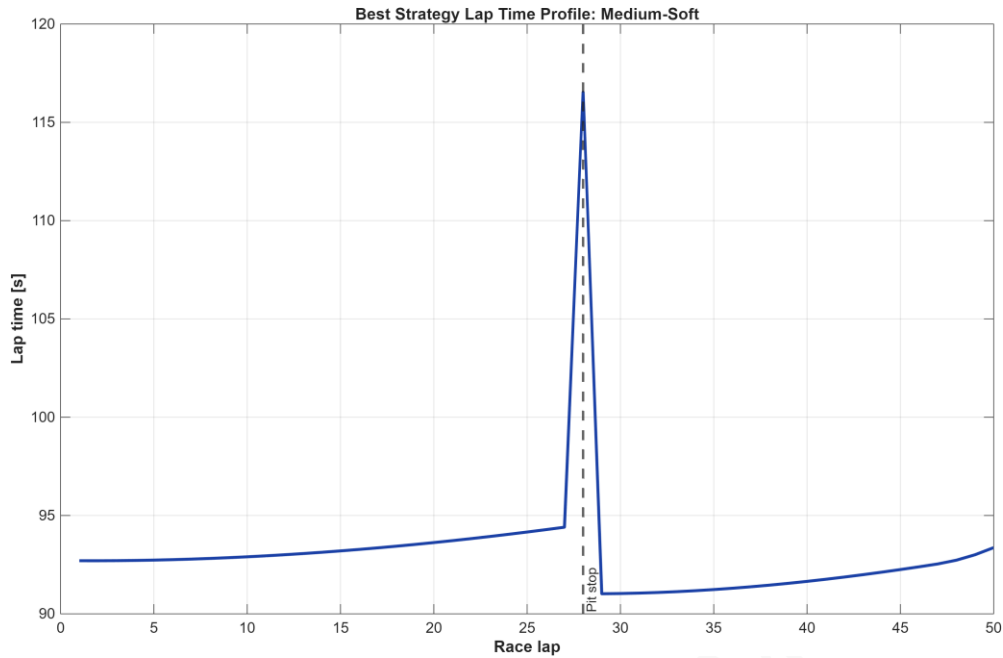


Figure 4. Lap time profile of the selected optimal strategy.

The lap time profile shows the evolution of race pace across both stints. The pit stop appears as a clear time loss peak, after which tyre age is reset and the second compound begins its own degradation cycle.

This figure helps visualize why the selected strategy performs best. The first stint is extended until the degradation penalty begins to outweigh the benefit of staying on track. After the pit stop, the second compound provides a more favourable performance profile for the remaining laps.

The model therefore identifies the optimal strategy by minimizing the total accumulated race time, rather than by selecting the compound with the fastest initial pace. This confirms that strategic performance depends on the interaction between tyre degradation, fuel burn-off, stint length and compound selection.

6. Discussion

6.1 Strategy Recommendation

Based on the total race time minimization, the recommended strategy is **Medium-Soft** as it achieves the lowest simulated total race time. However, the gap to Hard-Soft is very small (+0.48 s), which suggests that both strategies are competitive under the model assumptions. Under the assumptions of the model, this strategy (medium-soft) provides the best compromise between compound pace, degradation control, fuel-load sensitivity and pit stop time loss.

The first stint should be extended until lap 26 approximately (we can see it in the pit window optimization graph), as stopping earlier would not fully exploit the operating window of the starting compound. However, extending the stint beyond this point increases the accumulated degradation penalty and results in a higher total race time. Therefore, the selected pit lap represents the point where the benefit of fitting a new compound becomes greater than the advantage of staying out.

-Important: This recommendation should be interpreted under stable race conditions. It does not account for traffic, safety car periods, tyre temperature windows, overtaking difficulty or weather variation.

6.2 Technical Interpretation

The simulation confirms that the optimal strategy is driven by cumulative race time rather than peak compound performance. The Soft tyre provides the lowest initial lap time, but its higher degradation gradient and stronger sensitivity to fuel load reduce its efficiency when the stint is extended. In contrast, the Medium and Hard compounds generate a more stable degradation profile, which can make them more effective over longer race segments despite their slower baseline pace.

The optimum pit stop lap is reached when the marginal time loss from tyre ageing exceeds the expected benefit of continuing the stint. From a race engineering perspective, this represents the point where the degradation rate becomes strategically more expensive than the pit stop loss and the performance reset achieved with a new tyre set.

The fuel-load sensitivity correction is important because tyre degradation is not only a function of tyre age. A compound used in the first stint operates under higher fuel mass, higher vertical load and greater energy input, which increases thermal and mechanical stress. This makes early-stint degradation more severe and avoids unrealistic symmetry between strategy permutations such as Soft-Medium and Medium-Soft.

The tyre cliff penalty was included to represent the operational risk of running a compound beyond its estimated performance window. Without this constraint, the optimizer may select long stints on softer compounds because the degradation curve remains mathematically continuous.

6.3 Model Limitations

The model is intentionally simplified and does not include track evolution, tyre temperature windows, traffic losses, overtaking difficulty, safety car probability, weather variation, driver management or real tyre supplier data. In a trackside environment, these variables would be essential for a final strategy call.

However, within the scope of this project, the model provides a structured strategy evaluation framework by linking compound behaviour, stint length, fuel effect and pit stop timing into a single total race time calculation.

7. Future Improvements

Although the model provides a structured framework for one-stop strategy evaluation, several improvements could be introduced to increase its realism and strategic depth.

Potential developments include:

- Track evolution: include the effect of increasing grip as the circuit rubbers in during the race.
- Tyre temperature window: model warm-up, overheating and performance loss outside the optimal operating range.
- Traffic and overtaking losses: estimate the time lost when rejoining behind slower cars after the pit stop.
- Safety car scenarios: evaluate how neutralisations could shift the optimal pit window.
- Two-stop strategies: compare one-stop and two-stop race plans under different degradation levels.
- Real data calibration: use telemetry or simulator-derived lap times to adjust degradation coefficients and validate the model.

These improvements would make the model closer to a real race.

8. Conclusion

This project created a simplified MATLAB race strategy model assessing tyre degradation, fuel burn-off, fuel-load sensitivity, and pit stop timing's impact on total race time. The model evaluates various one-stop strategies to determine optimal pit stop laps by minimizing total race time. Results indicate that the ideal strategy balances stint length, degradation control, fuel load, and performance reset post-pit stop rather than just initial pace. Medium-Soft emerges as the preferred strategy, optimizing race conditions while managing heavy fuel loads initially and maximizing Soft's performance in lighter fuel conditions. Despite its simplifications, the model offers a structured framework for race strategy evaluation and supports performance analysis.